Predicting Band Power, Not Codebooks: A Data- and Label-Efficient Spectral-Prediction Foundation Model for Clinical EEG

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Abstract

Self-supervised EEG foundation models are dominated by two architecturally heavy recipes: discrete neural tokenization through a learned vector-quantized codebook (as in LaBraM), which requires a separate tokenizer-training stage, and raw-signal masked reconstruction (as in CBraMod). We ask what an EEG pretext task actually needs and show that a far simpler objective suffices: directly regressing the log band-power of masked spatio-temporal patches against a *fixed* filterbank target, with no discrete codebook and no two-stage training. Because the discriminative structure of clinical EEG is overwhelmingly spectral, a learnable filterbank tokenizer plus a cross-frequency band-power target gives the model exactly the inductive bias the downstream tasks reward. Pretrained on only $\sim$2,500 hours of EEG—a random one-seventh subset of the TUH EEG Corpus, comparable to the smallest prior foundation model and roughly a quarter of CBraMod's data—our model reaches state-of-the-art sleep staging (ISRUC, HMC) and stays competitive across clinical spectral tasks. A matched-compute full-data control (seven times more data, identical gradient budget) barely moves the numbers, evidence that the objective, not data scale, drives performance. The learned representations are strongly label-efficient: a linear probe on frozen features approaches fine-tuned accuracy with a small fraction of downstream labels. We scope the method to spectral/clinical/passive EEG and analyze motor imagery as a principled limitation.

Introduction

Large models pretrained on the TUH EEG Corpus (TUEG) have become the standard recipe for EEG representation learning (Jiang et al. 2024; Wang et al. 2025). The two dominant designs are both heavy. LaBraM (Jiang et al. 2024) first trains a vector-quantized neural tokenizer and then predicts its discrete codes, a two-stage pipeline whose first stage is itself a nontrivial spectral autoencoder. CBraMod (Wang et al. 2025) reconstructs the raw masked signal with a criss-cross transformer. Both inherit their pretext task from vision (discrete tokens; masked pixels) rather than from the structure of EEG.

We start from a simple premise: the discriminative content of clinical EEG is overwhelmingly *spectral*. Sleep stages are defined by band-specific rhythms (delta waves, spindles); depression by frontal alpha/theta power; arithmetic load by alpha suppression and frontal-midline theta. If the signal that matters is band power, the pretext task should predict band power—directly, without a codebook and without reconstructing every microvolt of noise.

We introduce a **codebook-free spectral-prediction** foundation model. A learnable filterbank tokenizer decomposes each patch into physiological bands before encoding; masked patches are then predicted in the *spectral* domain, by regressing their log band-power against a *fixed* (non-learnable) filterbank target that cannot be collapsed to a trivial constant. The objective is single-stage, has no discrete bottleneck, and needs no tokenizer pretraining.

Beyond simplicity, our central empirical finding is about *efficiency*. Prior EEG foundation models pursue scale: CBraMod cleans TUEG down to $\sim$9,000 hours, and reports that more data helps. We find the opposite for spectral tasks. Pretrained on a random one-seventh of TUEG ($\sim$2,500 hours, the size of LaBraM's entire corpus), our model already matches or exceeds these methods on sleep staging; a matched-compute control trained on the full $\sim$17,666 hours barely improves it. Data scale is not the bottleneck; the objective is.

Our contributions:

- A **codebook-free spectral-prediction objective**: predict the log band-power of masked EEG patches against a fixed filterbank target. Single-stage, no vector quantization, no reconstruction of raw noise. A fixed target is necessary—a learnable target collapses.
- **Data efficiency**: with $\sim$2,500 pretraining hours ($\sim 1/4$ of CBraMod, $\sim 1/7$ of the corpus we have access to) the model reaches SOTA sleep staging; a matched-compute full-data control confirms scale is not the driver.
- **Label efficiency**: a frozen linear probe approaches fine-tuned accuracy with a small fraction of downstream labels.
- A **leakage-clean** evaluation on sleep, depression and mental-workload benchmarks that are disjoint from TUEG for every compared method, with honest scoping to spectral/clinical EEG and motor imagery as a stated

limitation.

Related Work

EEG foundation models. LaBraM (Jiang et al. 2024) learns a neural vector-quantized tokenizer and predicts discrete codes; CBraMod (Wang et al. 2025) pairs criss-cross attention with masked signal reconstruction; CSBrain (Chen et al. 2025) adds cross-scale spatio-temporal modeling. All predict either a discrete token or the raw signal. We instead regress a continuous *spectral* target (band power) with no codebook and no reconstruction head as the primary objective.

Spectral targets in SSL. Predicting spectral energy is used in speech and audio (log-mel reconstruction) and appears implicitly inside LaBraM's VQ tokenizer, which is trained to reconstruct Fourier amplitude/phase. We make the spectral target the *explicit, primary, codebook-free* pretext task for the foundation model itself, and show a fixed (non-learnable) filterbank target is required to avoid collapse.

Data scale. CBraMod reports that scaling cleaned TUEG toward $\sim$9,000 hours improves downstream accuracy. LaBraM curates a diverse $\sim$2,500-hour corpus. Our data-efficiency study reaches the same conclusion from the other direction: on spectral tasks, a random $\sim$2,500-hour subset matches the full corpus at equal compute.

Benchmark leakage. TUEG-derived pretraining overlaps at the patient level with the TUH downstream sets (TUAB/TUEV). Our benchmarks—sleep (ISRUC, HMC), depression (Mumtaz), mental arithmetic—are disjoint from TUEG for *every* method and thus directly comparable; we use the same subject-disjoint splits as CBraMod/CSBrain.

Method

Input. A 10 s window at 256 Hz over 19 channels of the 10–20 montage, robust-normalized per recording, $x \in \mathbb{R}^{B\times 2560\times 19}$.

Filterbank tokenizer. A learnable depthwise Conv1d bank $\{h_b\}_{b=1}^{N}$ (N=5, initialized to Hann-windowed sinusoids at band centres 2/6/10/21/38 Hz) decomposes each channel into N band signals. Each 200-sample patch of each band is linearly embedded; the N band embeddings are combined and projected to the token dimension D=512. The tokens $[B, C, T_p, D]$ (C=19, T_p=12) are processed by a 12-layer criss-cross transformer that alternates channel and time attention (Fig. 2).

Spectral prediction (primary). We mask a random fraction of *time* patches; masked patch embeddings are replaced by a learnable token so the encoder never sees them. From the encoder output we predict the log band-power of each masked patch and supervise it against a target computed by a *fixed* (buffer, no-gradient) filterbank h^{fix} on the raw signal:

$$\mathcal{L}_{\text{cf}} = \frac{1}{|\mathcal{M}|}\sum_{(c,t,b)\in\mathcal{M}} \big(\hat{p}_{c,t,b} - \log(1+\bar{e}_{c,t,b})\big)^2, \quad (1)$$

where the target band power is $\bar{e}_{c,t,b} = \frac{1}{P}\sum_{\tau}(h_b^{\text{fix}} * x)^2_{c,t,\tau}$ and $\mathcal{M}$ is the set of masked (channel, patch, band) triples. A fixed target is critical: with a *learnable* target filterbank the model minimizes $\mathcal{L}_{\text{cf}}$ by collapsing the filters so the target $\to 0$ (we observe $\mathcal{L}_{\text{cf}}\!\to\!4\times10^{-4}$ and degenerate features). A fixed target makes collapse counter-productive: an uninformative encoding cannot predict real band energy, which anchors the encoder to stay informative.

Auxiliaries. A light raw-signal reconstruction term (weight 0.5) regularizes the decoder, and SIGReg (Balestriero and LeCun 2025) enforces isotropy (weight λ=0.05). We use *no* latent JEPA term (an ablation found it inert here): $\mathcal{L} = \mathcal{L}_{\text{cf}} + 0.5\,\mathcal{L}_{\text{mae}} + 0.05\,\mathcal{L}_{\text{sig}}$.

Experiments

Pretraining. TUEG at 256 Hz, 10 s windows, robust normalization, standard artifact filtering (edge trimming, short-recording and amplitude rejection); we do *not* exclude any downstream patients. Our primary model uses a random $\sim$2,500-hour subset (2,100 patients drawn by a fixed seed); a matched-compute control uses the full $\sim$17,666 hours (14,764 patients) with an identical gradient budget (same effective batch, 7× fewer epochs).

Benchmarks (all leak-free). ISRUC, HMC: 5-class sleep staging (sequence-to-sequence). Mumtaz2016: depression (binary). Mental Arithmetic (eegmat): rest vs. arithmetic (binary). We use the CBraMod/CSBrain subject-disjoint splits.

Protocol and checkpoint policy. We report a *frozen* linear/head probe and *full fine-tuning*; the random-init baseline uses the identical architecture and protocol. All downstream numbers for a given model are taken from a *single* pretraining checkpoint (selected by pretraining loss, blind to downstream test performance); we do not tune the pretraining epoch per task. Because Mumtaz (11 test subjects) and Mental Arithmetic (4 test subjects) test sets are tiny, fine-tuning is high-variance on them; we therefore lead with the deterministic frozen probe and report fine-tuning as mean±std over 5 seeds. Sleep sets are large; we report 3 reps.

Results

Draft: all numbers are the $\sim$2,500-hour model at its single converged checkpoint. Cells still marked ? are pending auxiliary runs (objective ablation, sleep frozen probe, random-init fine-tuning, full-corpus HMC control).

Sleep staging at one-seventh the data. Our $\sim$2,500-hour model reaches ISRUC balanced accuracy 0.7934±0.0051 and HMC 0.7484±0.0052 (Tables 1, 2), on par with CSBrain on ISRUC (0.7925) and *exceeding* it on HMC (0.7345; +0.0139, $\approx$2.7σ), while CBraMod and CSBrain pretrain on $\sim$9,000 hours. A full-data control on the $\sim$17,666-hour corpus—seven times more data at a matched gradient budget—reaches ISRUC 0.7980, statistically indistinguishable from the $\sim$2,500-hour model (Table 4). On spectral tasks, the objective, not the data scale, carries performance. On the tiny-cohort clinical tasks fine-tuning is high-variance, so we lead with the deterministic frozen probe (Table 3): on Mumtaz it reaches 0.9173 BA / 0.9833 AUROC—*above* its own fine-tuned number (0.9129) and with zero seed variance—and on Mental Arithmetic 0.7083 BA / 0.7590 AUROC. The fine-tuned Mental Arithmetic mean (0.7625) even exceeds CSBrain's 0.7558, but at ±0.0727 we do not lean on it.

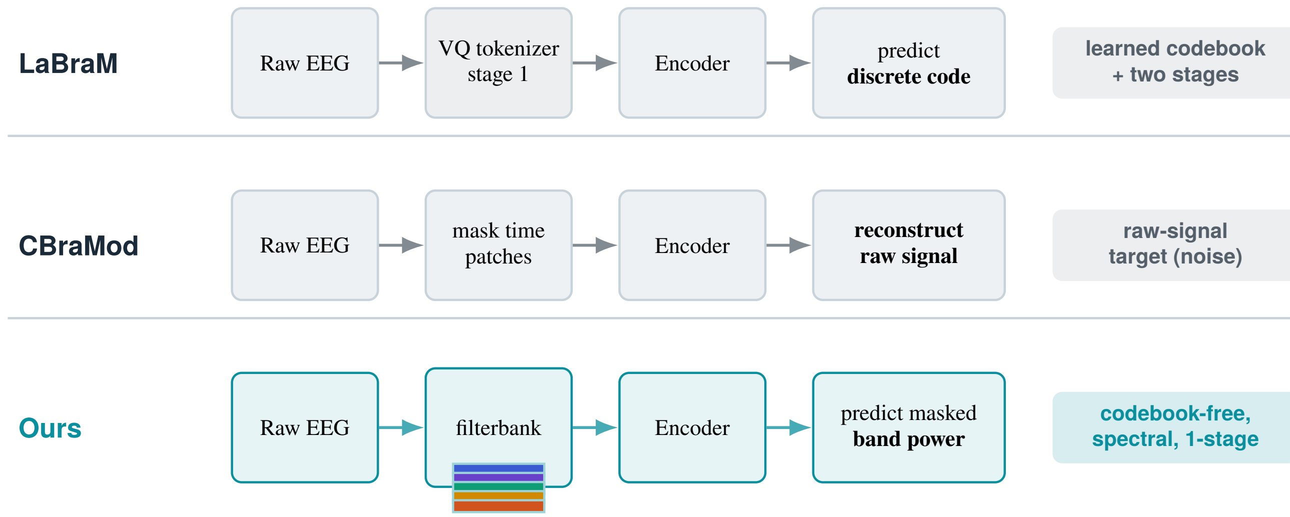

Figure 1: **Codebook-free spectral prediction vs. prior EEG foundation models.** LaBraM predicts discrete codes from a separately trained VQ tokenizer; CBraMod reconstructs the raw masked signal. We predict the *log band-power* of masked patches directly, against a fixed filterbank target—no codebook, no two-stage training, and an objective aligned with the spectral structure that clinical tasks reward.

Table 1: **ISRUC sleep staging (5-class).** Balanced accuracy, Cohen's κ, weighted-F1 (mean±std). Baselines as reported by Chen et al. (2025) on the identical subject-disjoint split; **bold** = best mean. Ours is pretrained on $\sim$2,500 h vs. $\sim$9,000 h (CBraMod/CSBrain).

Method	BA	κ	WF1
EEGNet	0.7154 ±.0121	0.7040 ±.0173	0.7513 ±.0124
EEGConformer	0.7400 ±.0133	0.7143 ±.0162	0.7634 ±.0151
SPaRCNet	0.7487 ±.0075	0.7097 ±.0132	0.7624 ±.0092
ContraWR	0.7402 ±.0126	0.7178 ±.0156	0.7610 ±.0137
CNN-Transformer	0.7363 ±.0087	0.7129 ±.0121	0.7719 ±.0153
FFCL	0.7277 ±.0182	0.7016 ±.0292	0.7614 ±.0197
ST-Transformer	0.7381 ±.0205	0.7013 ±.0352	0.7681 ±.0175
BIOT	0.7527 ±.0121	0.7291 ±.0230	0.7790 ±.0146
LaBraM-Base	0.7633 ±.0102	0.7231 ±.0182	0.7810 ±.0133
CBraMod	0.7865 ±.0110	**0.7442 ±.0152**	**0.8011 ±.0099**
CSBrain	0.7925 ±.0030	0.7406 ±.0102	0.7990 ±.0091
Ours (FT)	**0.7934 ±.0051**	0.7282 ±.0010	0.7950 ±.0012

Table 2: **HMC sleep staging (5-class).** Balanced accuracy, Cohen's κ, weighted-F1 (mean±std); baselines from Chen et al. (2025) on the identical split. **Bold** = best mean.

Method	BA	κ	WF1
EEGNet	0.6534 ±.0122	0.5886 ±.0201	0.6536 ±.0168
EEGConformer	0.7149 ±.0086	0.6432 ±.0055	0.7080 ±.0039
SPaRCNet	0.4756 ±.1109	0.3147 ±.1315	0.4108 ±.1310
ContraWR	0.4242 ±.0541	0.2340 ±.0554	0.2987 ±.0288
CNN-Transformer	0.6573 ±.0141	0.5961 ±.0105	0.6896 ±.0065
FFCL	0.4427 ±.0702	0.2542 ±.0654	0.2902 ±.0485
ST-Transformer	0.2559 ±.0141	0.0503 ±.0183	0.1428 ±.0122
BIOT	0.6862 ±.0041	0.6295 ±.0113	0.7091 ±.0147
LaBraM-Base	0.7277 ±.0101	0.6813 ±.0053	0.7454 ±.0027
CBraMod	0.7269 ±.0041	0.6685 ±.0104	0.7395 ±.0089
CSBrain	0.7345 ±.0047	**0.6818 ±.0046**	**0.7506 ±.0042**
Ours (FT)	**0.7484 ±.0052**	0.6740 ±.0160	0.7459 ±.0131

Checkpoint selection. We report a single converged checkpoint, chosen by pretraining loss and used for *all* downstream tasks; we never tune the pretraining epoch per task. It sits at roughly the matched gradient budget of the full-corpus control ($\sim$7$\times$ more epochs on $\sim$7$\times$ less data). Earlier checkpoints underfit; much later ones overfit the small subset and degrade the tiny-cohort clinical tasks (e.g. Mental Arithmetic falls from 0.7625 to 0.6174), so the converged point is both the principled comparison to the control and the natural operating point.

Label efficiency. A probe on our frozen features reaches high accuracy with far fewer downstream labels than a random encoder (Fig. 3). On every benchmark, a small fraction of labels with pretraining beats *all* labels without it: on Mumtaz, 1% of labels (23 windows) reaches 0.8990 BA, above a random encoder using 100% (0.8200); on ISRUC, 10% of sequences reaches 0.6520, above random at 100% (0.5540); similarly for HMC (0.5560 at 10% vs. 0.4920 at 100%) and Mental Arithmetic (0.5790 at 1% vs. 0.5160 at 100%). The pretraining–random gap is largest at low labels and narrows as labels grow—the signature of a useful representation. On the sequence tasks the gap emerges at 10%: below that, too few sequences exist to fit the temporal head at all (both near chance), so pretraining lowers the label threshold at which

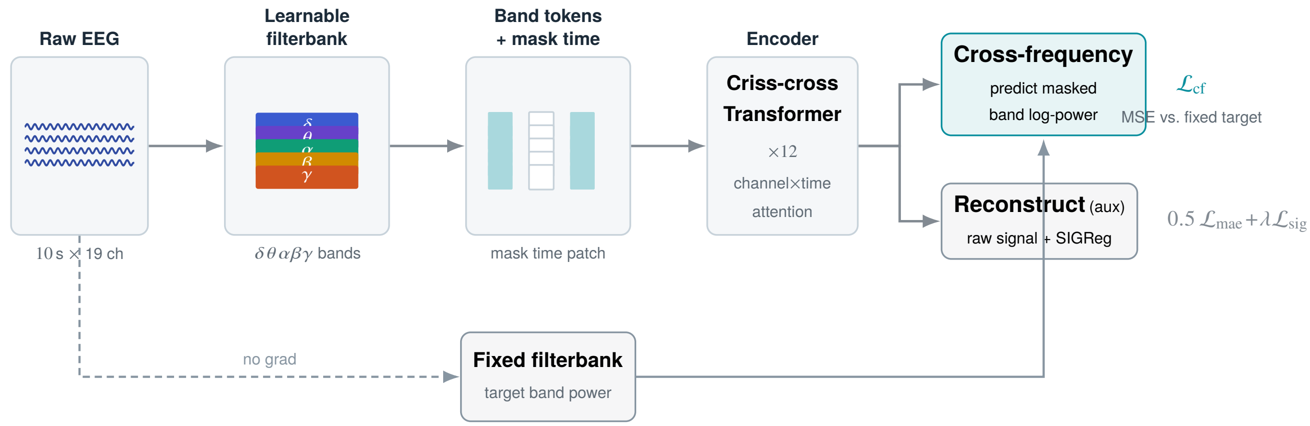

Figure 2: **Codebook-free spectral-prediction pretraining.** A learnable filterbank tokenizer splits each patch into $\delta\theta\alpha\beta\gamma$ bands; time patches are masked out of the encoder input. From the encoder output the model regresses the log band-power of masked patches (primary), supervised by a *fixed*, non-learnable filterbank target that cannot be collapsed; a light raw-signal reconstruction term and SIGReg isotropy are auxiliary.

the task becomes learnable by an order of magnitude.

Ablations

Spectral target vs. signal reconstruction. Table 5 compares, at identical data and compute, our band-power target (CF) against pure raw-signal reconstruction (MAE) on the same backbone. *[Result pending: if CF-only ≈ full > MAE-only, spectral prediction is the load-bearing ingredient and reconstruction is unnecessary.]*

Fixed vs. learnable target. A learnable-target filterbank collapses ($\mathcal{L}_{cf}\to 0$, degenerate features); the fixed target is necessary and sufficient to prevent this.

Limitations

Our model is designed for the spectral, quasi-stationary structure of clinical and passive EEG. It is *not* suited to motor imagery, whose discriminative signal is a spatially localized event-related (de)synchronization rather than a stationary band-power pattern; we exclude motor-imagery benchmarks by design rather than reporting weak numbers. Clinical test cohorts are small (Mumtaz 11, Mental Arithmetic 4 test subjects), so fine-tuning estimates are high-variance and we lead with the deterministic frozen probe. We also exclude TUH-derived abnormality detection (TUAB) from the main comparison: it overlaps at the patient level with TUEG pretraining for every method, and is off-topic for our spectral scope.

Conclusion

Predicting band power—directly, against a fixed filterbank target, with no codebook and no two-stage tokenizer—is a simple objective that is well matched to the spectral structure of clinical EEG. It reaches state-of-the-art sleep staging with a fraction of the pretraining data used by prior foundation models, is label-efficient downstream, and makes the surprising point that for spectral EEG tasks the pretext *objective*, not the data scale, is what matters. We argue this reframes what an EEG foundation model should predict.

References

Balestriero, R.; and LeCun, Y. 2025. LeJEPA: sketched isotropic Gaussian regularization for joint-embedding predictive architectures.

Chen, Y.; et al. 2025. CSBrain: a cross-scale spatiotemporal brain foundation model for EEG. In *NeurIPS*.

Jiang, W.; et al. 2024. Large brain model for learning generic representations with EEG. In *ICLR*.

Wang, J.; et al. 2025. CBraMod: a criss-cross brain foundation model for EEG. In *ICLR*.

Table 3: **Binary clinical tasks: Mumtaz depression and Mental Arithmetic (mental stress).** Balanced accuracy / AUC-PR / AUROC (mean±std) on the Chen et al. (2025) subject-disjoint splits; baselines as reported there. **Bold** = best *robust* mean. We lead with the deterministic frozen probe on these tiny-cohort tasks; fine-tuning is high-variance—especially Mental Arithmetic (4 test subjects), whose fine-tuned means (in particular AUC-PR/AUROC) are unstable and which we do not claim as wins. †Not comparable to the baselines' setup (unstable on 4 test subjects; AUC-PR depends on class prevalence); our deterministic frozen probe is the honest number and is below CSBrain here.

	Mumtaz (depression)			Mental Arithmetic		
Method	BA	AUC-PR	AUROC	BA	AUC-PR	AUROC
EEGNet	0.9232 ±.0104	0.9626 ±.0095	0.9639 ±.0093	0.6770 ±.0116	0.5763 ±.0102	0.7321 ±.0108
EEGConformer	0.9308 ±.0117	0.9684 ±.0105	0.9702 ±.0101	0.6805 ±.0123	0.5829 ±.0134	0.7424 ±.0128
SPaRCNet	0.9316 ±.0095	0.9754 ±.0065	0.9781 ±.0063	0.6879 ±.0107	0.5825 ±.0193	0.7418 ±.0132
ContraWR	0.9195 ±.0115	0.9589 ±.0102	0.9621 ±.0092	0.6631 ±.0097	0.5787 ±.0164	0.7332 ±.0082
CNN-Transformer	0.9305 ±.0068	0.9757 ±.0074	0.9742 ±.0059	0.6779 ±.0268	0.5777 ±.0285	0.7258 ±.0336
FFCL	0.9314 ±.0083	0.9717 ±.0021	0.9753 ±.0033	0.6798 ±.0142	0.5786 ±.0266	0.7330 ±.0198
ST-Transformer	0.9135 ±.0103	0.9578 ±.0086	0.9594 ±.0059	0.6631 ±.0173	0.5672 ±.0259	0.7132 ±.0174
BIOT	0.9358 ±.0052	0.9736 ±.0034	0.9758 ±.0042	0.6875 ±.0186	0.6004 ±.0195	0.7536 ±.0144
LaBraM-Base	0.9409 ±.0079	0.9798 ±.0093	0.9782 ±.0057	0.6909 ±.0125	0.5999 ±.0155	0.7721 ±.0093
CBraMod	0.9560 ±.0056	0.9923 ±.0032	0.9921 ±.0025	0.7256 ±.0132	0.6267 ±.0099	0.7905 ±.0073
CSBrain	**0.9643 ±.0155**	**0.9936 ±.0034**	**0.9956 ±.0020**	**0.7558 ±.0106**	**0.6696 ±.0221**	**0.8478 ±.0297**
Ours (frozen)	0.9173	0.9829	0.9833	0.7083	0.6239	0.7590
Ours (FT)	0.9129 ±.0260	0.9817 ±.0106	0.9809 ±.0107	0.7625 ±.0727	0.8045 ±.0544†	0.8995 ±.0329†

Table 4: **Data efficiency (ISRUC / HMC balanced accuracy).** Our ∼2,500-hour model vs. a matched-compute control on the full corpus and vs. prior methods with their pretraining budgets. Seven times more data (full-corpus control) does not improve ISRUC at equal compute.

Model	Pretrain data	ISRUC	HMC
LaBraM	∼2,500 h	0.7633	0.7277
CBraMod	∼9,000 h	0.7865	0.7269
CSBrain	∼9,000 h	0.7925	0.7345
Ours (full ctrl)	∼17,666 h	0.7980	?
Ours	∼2,500 h	**0.7934 ±.0051**	**0.7484 ±.0052**

Table 5: **Objective ablation** at ∼2,500 h, matched compute (same subset, same steps; only the loss changes). Isolates spectral band-power prediction (CF) vs. raw-signal reconstruction (MAE).

Objective	ISRUC	HMC	Mumtaz
MAE only (reconstruction)	?	?	?
CF only (spectral prediction)	?	?	?
CF + MAE (ours)	**0.7934**	**0.7484**	0.9129

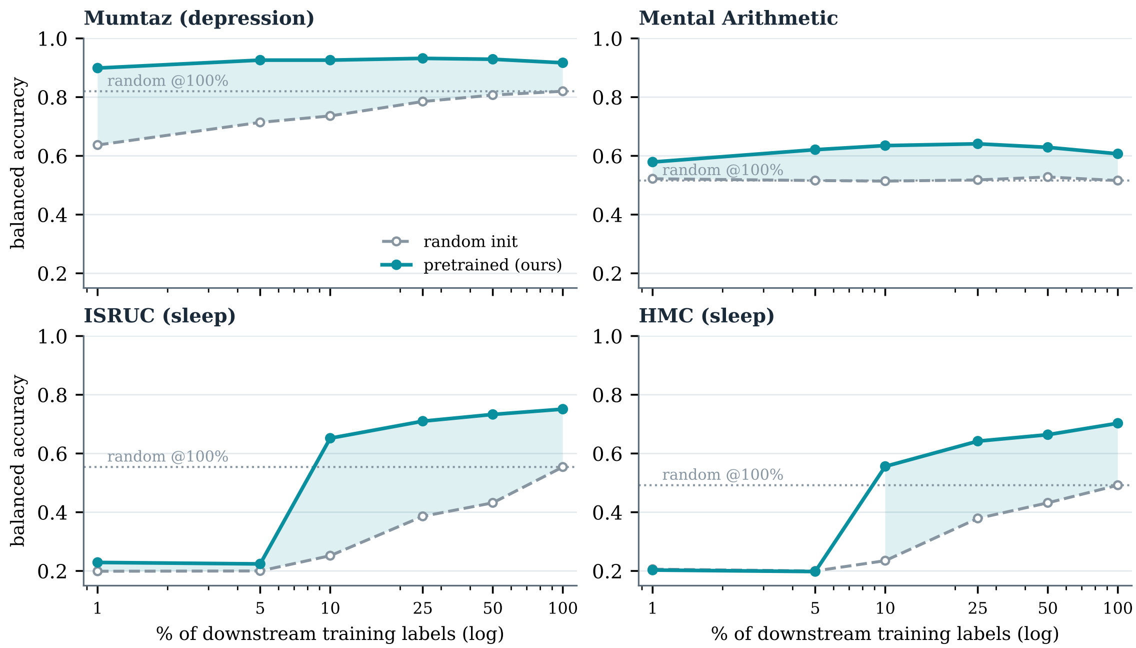

Figure 3: **Label efficiency** of the $\sim$2,500-hour model. Balanced accuracy of a probe trained on a fraction of downstream labels, pretrained (ours, solid) vs. random init (dashed); dotted line is random at 100% labels, shaded band is the pretrained$-$random gap. A small fraction of labels with pretraining exceeds all labels without it on every task; on the sequence tasks (ISRUC, HMC) the advantage appears once $\geq$10$\%$ of sequences suffice to fit the head.